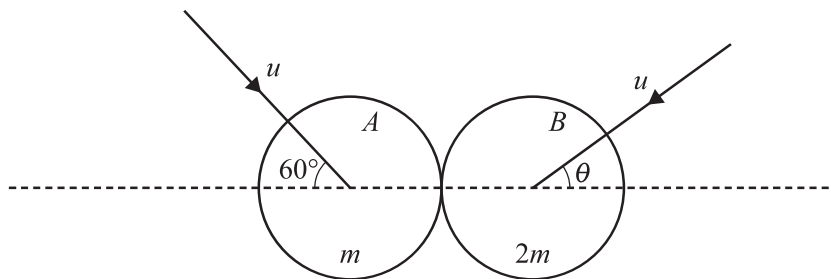


1



Two uniform smooth spheres A and B of equal radii have masses m and $2m$ respectively. The two spheres are moving with equal speeds u on a smooth horizontal surface when they collide. Immediately before the collision, A 's direction of motion makes an angle of 60° with the line of centres, and B 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e .

After the collision, the component of the velocity of A along the line of centres is v and B moves perpendicular to the line of centres. Sphere A now has twice as much kinetic energy as sphere B .

- (a) Show that $v = \frac{1}{2}u(4 \cos \theta - 1)$. [1]

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- (b) Find the value of $\cos \theta$. [4]

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(c) Find the value of e . [2]

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